

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Class Test

Code of Conduct:

I Cheredla Jasvina/192324026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	
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Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Scenario: A drone is used to capture aerial images of a farmland to identify pest affected crop regions. Due to sensor limitations, wind movement, and changing sunlight, the captured images contain noise, shadows, and uneven illumination. Pest affected areas usually show visible texture and color variations compared to healthy crop regions.

Answer the following questions:

a) Preprocessing Pipeline — 4 Marks

Propose a suitable preprocessing pipeline to improve the quality of the farmland images before analysis. Your answer should include methods for noise reduction, illumination correction, contrast enhancement, and normalization. Justify the purpose of each step.

Answer:

Image preprocessing is the first step in image analysis. It removes unwanted information from the image and improves its quality so that feature extraction and classification become more accurate.

Step 1: Noise Reduction

Method: Gaussian Filter or Median Filter

Noise is the unwanted random variation in pixel values that appears because of camera sensor limitations, drone vibration, or environmental conditions like wind. If noise is not removed, the algorithm may mistake noisy pixels for pest-damaged regions.

A Gaussian Filter smooths the image by replacing each pixel with the weighted average of its neighboring pixels. It is mainly used for removing Gaussian noise while preserving the overall structure of the image.

A Median Filter replaces each pixel with the median value of its neighboring pixels. It is very effective in removing salt-and-pepper noise without blurring the edges.

Example:

Suppose the pixel values are:

120 122 121

121 255 122

120 121 122

The value **255** is a noisy pixel.

After applying a Median Filter:

120 121 121

121 121 122

120 121 122

The noisy pixel is replaced with **121**, giving a smoother image.

Purpose:

- Removes unwanted sensor noise.
- Produces a cleaner image.
- Prevents false detection of pests.

Step 2: Illumination Correction

- **Method:** CLAHE (Contrast Limited Adaptive Histogram Equalization)
- Drone images are captured under natural sunlight. Some parts of the farmland may receive direct sunlight while others remain under shadows. As a result, one part of the image appears very bright and another appears very dark. This uneven illumination makes pest detection difficult.
- CLAHE divides the image into small regions and improves the brightness and contrast of each region separately. Unlike normal Histogram Equalization, CLAHE prevents over-enhancement and preserves image details.

Example:

Before CLAHE:

- Left side of the field = Bright
- Right side = Dark due to cloud shadow

After CLAHE:

- Both regions have nearly equal brightness.
- Pest-damaged leaves become clearly visible.

Purpose:

- Removes the effect of shadows.
- Corrects uneven lighting.
- Makes all crop regions uniformly visible.

Step 3: Contrast Enhancement

Method: Contrast Stretching or CLAHE

- Contrast is the difference between light and dark pixel values. If the contrast is low, healthy and diseased crops may appear almost identical.
- Contrast enhancement increases the difference between these regions so that pest damage becomes easier to identify.

Example:

- Before enhancement:
- Healthy Crop = 120
- Diseased Crop = 130
- Difference = **10**
- After enhancement:

- Healthy Crop = 80
- Diseased Crop = 170
- Difference = **90**
- Now the diseased crop is clearly distinguishable from the healthy crop.

Purpose:

- Improves visibility of pest-affected regions.
- Makes texture and color differences more noticeable.
- Helps feature extraction algorithms detect damage accurately.

Step 4: Normalization

- **Method:** Min-Max Normalization
- Different drone images may have different brightness ranges due to varying camera settings and lighting conditions. Normalization converts all pixel values into a common range, usually **0–1** or **0–255**.

The formula is:

$$I_{new} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

where:

I = Original pixel value

I_{min} = Minimum pixel value

I_{max} = Maximum pixel value

Example:

Original values:

80 120 200

Normalized values:

0.2 0.5 1.0

- All images now have the same intensity range.

Purpose:

- Standardizes image intensity.
- Makes different images comparable.
- Improves the performance of machine learning and computer vision algorithms.

- **Complete Preprocessing Pipeline**

Drone Image



Noise Reduction

(Gaussian Filter / Median Filter)



Illumination Correction

(CLAHE)



Contrast Enhancement
(Contrast Stretching / CLAHE)



Normalization
(Min-Max Normalization)



Preprocessed Image

This preprocessing pipeline produces a clean, bright, and standardized image that is ready for feature extraction.

b) Feature Extraction — 3 Marks

Choose an appropriate feature extraction method to identify texture changes caused by pest damage. Explain why the selected method is suitable for detecting affected crop regions.

Answer :

Selected Method: Local Binary Pattern (LBP)

Feature extraction is the process of identifying useful characteristics from an image. Since pest-damaged crops mainly differ in texture, the Local Binary Pattern (LBP) method is the most suitable choice.

LBP is a texture descriptor that compares every neighboring pixel with the center pixel.

Working of LBP

1. Select the center pixel.
2. Compare each neighboring pixel with the center pixel.
3. If the neighboring pixel is greater than or equal to the center pixel, assign 1.
4. Otherwise, assign 0.
5. Read the binary values clockwise to form a binary number.
6. Convert this binary number into a decimal value, which becomes the texture feature.

Example:

Center pixel = 50

Neighbourhood:

55 52 40

60 50 45

48 70 35

Comparison:

$55 \geq 50 \rightarrow 1$

$52 \geq 50 \rightarrow 1$

$40 < 50 \rightarrow 0$

$60 \geq 50 \rightarrow 1$

$45 < 50 \rightarrow 0$

$48 < 50 \rightarrow 0$

$70 \geq 50 \rightarrow 1$

$35 < 50 \rightarrow 0$

Binary Pattern:

11010010

This binary pattern represents the texture around the center pixel.

Why LBP is Suitable for Pest Detection

Healthy crops usually have smooth and uniform leaf textures, whereas pest-damaged crops contain holes, spots, discoloration, and rough surfaces. LBP captures these small texture variations very effectively.

Advantages of LBP:

- Excellent for texture analysis.
- Detects small local texture changes.
- Computationally simple and fast.
- Works well even when there are small illumination changes.
- Suitable for large aerial farmland images captured by drones.

Therefore, LBP accurately distinguishes pest-affected regions from healthy crop regions.

c)Rotation and Scale Variations — 3 Marks

Drone images may be captured from different heights and angles. Discuss how rotation and scale variations can affect feature detection. Suggest suitable methods to make the feature extraction process more robust.

Answer :

Drone images may be captured from different heights and angles, causing rotation and scale variations.

Effect of Rotation:

- Changes the orientation of crop features.
- Can reduce feature matching accuracy.

Effect of Scale:

- Changes the size of crop regions.
- Small pest details may become difficult to detect.

Methods to Improve Robustness:

- SIFT (Scale-Invariant Feature Transform): Detects features that are invariant to both rotation and scale.
- ORB (Oriented FAST and Rotated BRIEF): Faster than SIFT and robust to rotation.
- Image Pyramid: Detects features at multiple image scales, improving detection at different drone heights.

Conclusion:

Using SIFT, ORB, and Image Pyramid makes feature extraction robust to rotation and scale variations, resulting in more accurate pest detection.

Question 2: Stereo Vision and Motion Estimation

Scenario: A self-driving car is equipped with stereo cameras to navigate through a busy street. The system must estimate the distance of nearby vehicles and pedestrians in real time. It should also track moving objects accurately to avoid collisions, especially when pedestrians or vehicles change speed suddenly.

Answer the following questions:

a) Stereo Disparity and Depth Estimation — 4 Marks

Explain how stereo cameras use the difference between left and right image views to estimate the depth of objects. Discuss how disparity changes for nearby and distant objects.

Answer :

Stereo vision uses two cameras (left and right) placed a fixed distance apart, similar to human eyes. Since the cameras capture the same scene from slightly different viewpoints, the position of an object differs in the two images. This difference is called disparity.

The system finds the corresponding point in both images and calculates the disparity. Using the disparity, camera distance (baseline), and focal length, it estimates the object's depth.

Depth Formula:

$$\text{Depth}(Z) = \frac{f \times B}{d}$$

Where:

- f = Focal length of the camera
- B = Distance (baseline) between the two cameras
- d = Disparity

Disparity for Different Objects:

- Nearby objects: Have large disparity because they appear at very different positions in the two images. Hence, the estimated depth is small (object is close).
- Distant objects: Have small disparity because their positions are almost the same in both images. Hence, the estimated depth is large (object is far away).

Conclusion:

Stereo cameras estimate object distance by measuring disparity. Large disparity indicates nearby objects, while small disparity indicates distant objects

b) Optical Flow for Motion Tracking — 3 Marks

If a pedestrian suddenly changes speed or direction, explain how optical flow helps in tracking the pedestrian's motion between consecutive video frames.

Answer :

Optical flow is a technique used to estimate the motion of objects between consecutive video frames by tracking the movement of pixels.

If a pedestrian suddenly changes speed or direction, optical flow detects the change by comparing the pedestrian's position in successive frames. It calculates the direction and speed of movement, allowing the self-driving car to continuously track the pedestrian.

Advantages:

- Tracks moving pedestrians and vehicles accurately.
- Detects sudden changes in speed and direction.
- Helps the vehicle predict motion and avoid collisions.

Conclusion:

Optical flow continuously estimates the motion of objects, enabling safe navigation in dynamic traffic environments.

c) Errors and Mitigation Techniques — 3 Marks

Identify possible sources of error in depth estimation and motion tracking, such as camera misalignment, poor lighting, occlusion, fast motion, or textureless regions. Suggest suitable techniques to reduce these errors.

Answer :

Possible Sources of Error

- **Camera misalignment:** Produces incorrect disparity and depth estimation.
- **Poor lighting:** Makes object detection and matching difficult.
- **Occlusion:** Parts of an object may be hidden by another object.
- **Fast motion:** Causes motion blur, reducing tracking accuracy.
- **Textureless regions:** Lack of features makes matching difficult.

Techniques to Reduce Errors

- **Camera Calibration:** Corrects camera alignment and lens distortion.
- **Image Enhancement (CLAHE/Histogram Equalization):** Improves visibility in poor lighting.
- **Feature-based methods (SIFT/ORB):** Improve matching in difficult regions.
- **Kalman Filter:** Predicts object movement during fast motion or temporary occlusion.
- **High-frame-rate cameras:** Reduce motion blur and improve tracking.

Conclusion:

Using camera calibration, image enhancement, feature extraction methods, Kalman filtering, and high-frame-rate cameras improves the accuracy of depth estimation and motion tracking in self-driving cars.